

Module 13: How Populations Evolve - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. Microevolution is defined as:

- A. the origin of entirely new kingdoms of life
- B. a change in allele frequencies in a population over generations
- C. a change in a single individual's body
- D. the extinction of every species
- Answer: B
- Why: Microevolution is a shift in allele frequencies within a population over time.

2. Genetic drift has its strongest effect in:

- A. small populations
- B. very large populations
- C. populations with no variation
- D. populations that never reproduce
- Answer: A
- Why: Random sampling changes allele frequencies most strongly when the population is small.

3. A few birds colonize a new island and start a population with limited variation. This reduced variation is due to:

- A. natural selection alone
- B. gene flow between islands
- C. mutation pressure
- D. the founder effect
- Answer: D
- Why: The founder effect is genetic drift that occurs when a new population begins from only a few individuals.

4. The Hardy-Weinberg principle is useful mainly because it:

- A. proves evolution never happens
- B. measures average body size
- C. provides a non-evolving baseline to compare real populations against
- D. edits alleles directly
- Answer: C
- Why: Hardy-Weinberg describes allele frequencies with no evolution. Deviations from it signal that evolutionary forces are at work.